

Workload-Driven Custom RISC-V Instruction Extensions for Edge Video Anomaly Detection

Design Review

Member 1 Member 2 Member 3 Member 4

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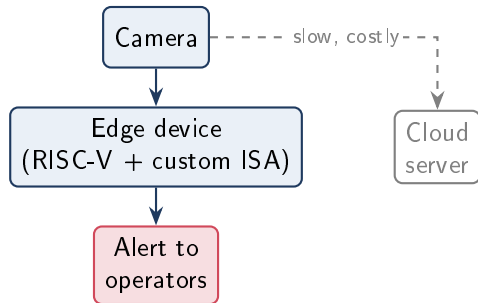
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Introduction: Why Edge AI for Surveillance?

- Metro / CCTV networks need **continuous, real-time** anomaly detection: violence, falls, crowd panic, fire, abandoned luggage.
- Sending every frame to a server costs **latency, bandwidth and privacy**.
- **Edge inference**: run the neural network on a small device next to the camera.
- Edge CPUs are slow at the core CNN operation, the **multiply-accumulate (MAC)**, which runs millions of times per frame.



Introduction: RISC-V and the Shakti C-Class Core

RISC-V

- Open, royalty-free instruction set architecture (ISA)
- Opcode space reserved for **custom instructions** (custom-0..3)
- Lets us add hardware operations tailored to a workload

Shakti C-Class (IIT Madras)

- 64-bit **RV64IMAFDC**, in-order, 6-stage pipeline
- Hardware FPU (Berkeley HardFloat), caches, MMU, AXI4 bus
- Written in **Bluespec SystemVerilog**, a **soft core** (can be simulated or put on an FPGA)
- Chosen over PicoRV32: a real pipeline, Linux-capable, and extensible

Simulation flow: BSV $\rightarrow$ bsc $\rightarrow$ Verilog $\rightarrow$ Verilator $\rightarrow$ cycle-accurate model bin/out

Problem Definition

- General-purpose embedded cores execute CNN kernels with scalar instructions: each MAC takes **11–18 cycles** on baseline Shakti.
- One MobileNetV2 + GRU inference (8 frames) takes **31.9 billion cycles** on baseline Shakti.
- On the existing Kria KV260 prototype (ARM + FPGA accelerator), **~82% of the frame time** is the ARM CPU reshuffling data; the accelerator itself uses only **~9%**.

Problem Statement

Design **workload-driven custom RISC-V instructions**, integrate them into the Shakti C-Class core, and deploy the core with an FPGA accelerator to speed up **edge video anomaly detection**, measured against an unmodified baseline.

Objectives

- ➊ **Profile** the target workloads (MobileNetV2 + GRU, then TinyMetro-Net) on PC *and* on the Shakti core itself.
- ➋ **Identify** the dominant kernels and the real hardware bottleneck.
- ➌ **Design** custom instructions in RISC-V custom opcode space.
- ➍ **Implement** them in the Shakti decoder and floating-point unit (RTL).
- ➎ **Verify** bit-exact results and backward compatibility (official riscv-tests).
- ➏ **Measure** speedup using hardware counters (mcycle, minstret).
- ➐ **Deploy** Shakti on the Kria KV260 FPGA in place of the ARM core, driving the INT8 accelerator.
- ➑ **Compare** edge (Shakti + accelerator) with server-based inference.

Functional & Non-Functional Requirements

Functional

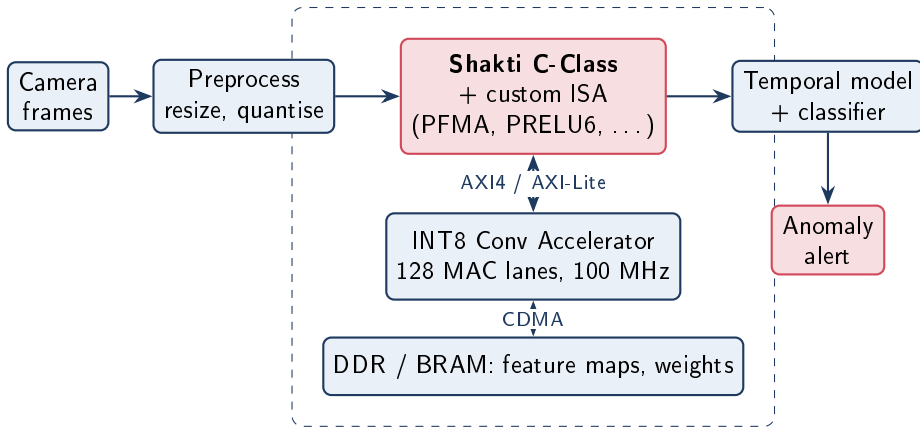
- Decode and execute new custom-opcode instructions
- Results **bit-identical** to standard RISC-V code
- Standard RV64IMAFDC behaviour unchanged
- Per-kernel cycle and instruction measurement
- Run the CNN kernels (conv, depthwise, pointwise, ReLU) on the core
- Interface with the FPGA accelerator over AXI4 (*planned*)

Non-Functional

- **Performance:** $\geq 1.5 \times$ kernel speedup
- **Correctness:** riscv-tests (rv64uf-fmadd) pass
- **Low overhead:** small hardware additions
- **Reproducibility:** open-source flow (bsc, Verilator, GCC 14)
- **Edge constraints:** limited memory (16 KB D-cache), real-time frame rate
- **Maintainability:** documented, logged design decisions

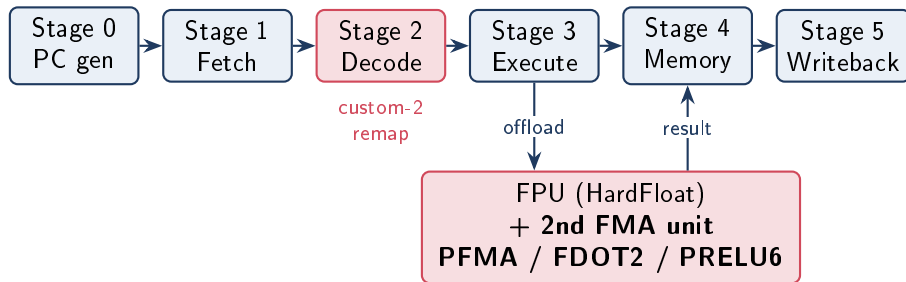
Overall System Architecture (Target Edge System)

Kria KV260 FPGA (programmable logic)



Shakti replaces the ARM Cortex-A53 as the host that drives the accelerator and runs the non-convolution work.

Architecture: Inside the Shakti C-Class Pipeline



- **Modified**: `decoder.bsv`, `decoder.defines`, `fpu_hardfloat.bsv`
- **Unchanged**: `fetch`, `scoreboard`, `forwarding`, `memory`, `writeback`. The new instructions reuse the existing `FLOAT` path.

Module 1: Workload Profiling (PC → Shakti)

PyTorch on a PC (MobileNetV2 + GRU)

Operation	MACs / frame
Pointwise 1×1	267.9 M
Depthwise 3×3	20.7 M
First conv 3×3	10.8 M
GRU (8 steps)	9.4 M

Measured on baseline Shakti (1 inference)

Operation	cyc/MAC	Share
Pointwise	11.7–13.9	83.8%
Depthwise	16.7	8.7%
First conv	17.7	4.8%
ReLU6	14.3/elem	2.2%
GRU	12.0	0.4%
Total	31.9 B cyc	

Finding

Pointwise convolution dominates on the real target. The FP32 data type rules out an INT8-only design, and Shakti already has `fmadd.s`, so a scalar custom MAC would add nothing.

Module 2: Custom Instruction Design

Instruction	fmt	Semantics	FPU op
FDOT2.S	00	$fd = a_1 b_1 + (a_0 b_0 + c)$, 2 FMAs <i>in sequence</i>	0110
PFMA.S	10	$fd = [x_1 w + c_1 \mid x_0 w + c_0]$, 2 FMAs <i>in parallel</i>	0111
PRELU6.S	01	$fd = [\text{clamp}_{0..6}(x_1) \mid \text{clamp}_{0..6}(x_0)]$, 1 cycle	0101

- All three use the **custom-2** opcode (0x5B) with the R4 layout; fmt bits [26:25] select the instruction.
- Two FP32 values are **packed** into one 64-bit FP register (FLEN = 64).
- **Design path:** FDOT2 halved the instruction count but gave only 1.07–1.13 $\times$. That revealed the real bottleneck: the FPU accepts **one operation at a time** (about 10-cycle FMA latency). $\Rightarrow$ PFMA adds a **second FMA unit** so two MACs overlap.

Module 3: Hardware Integration (RTL)

- 1 **Decoder remap**: rewrite custom-2 into an internal FMADD form, so the existing decode, scoreboard and writeback are reused unchanged.
- 2 **Own FPU opcode** (fn): the R4 f7 field holds register data, not instruction bits.
- 3 **Raw operands**: bypass NaN-boxing for packed values.
- 4 **Second FMA unit** spfma2 for PFMA lane 1.
- 5 **PRELU6**: integer compares on IEEE bits, 1 cycle.

```
// decoder.bsv (fn_decode)
if (inst_in[6:0] == 'b1011011)
  inst = {inst_in[31:27], 2'b10,
          inst_in[24:7], 7'b1000011};

// fpu_hardfloat.bsv (rule start)
if (opcode == 4'b0111) begin // PFMA
  spfma.request (0, r1[31:0],
                 r2[31:0], r3[31:0], f3);
  spfma2.request(0, r1[63:32],
                r2[31:0], r3[63:32], f3);
  rg_pfma <= True;
end
```

Module 4: Verification & Benchmarking

Verification

- Directed test: FDOT2 of $[2, 1.5] \cdot [4, 3] + 0.5 = 13.0$ ✓
- Every benchmark checks **bit-exact** equality with the baseline C code
- Official `rv64uf/fmadd.S` passes after each change
- Pass/fail read from `tohost`; traces in `rtl.dump`

Bugs found & fixed

- Stale decoder build (illegal-instruction trap)
- NaN-boxing turned packed inputs into NaN
- Wrong source for the FPU `f7` field
- BSC rule conflict: request and response in one rule

Benchmark flow: C kernel $\rightarrow$ RISC-V GCC $\rightarrow$ elf2hex $\rightarrow$ simulated Shakti $\rightarrow$ mcycle/minstret $\rightarrow$ report

Module 4: Measured Results (all bit-exact)

Kernel	Baseline	Custom	Speedup
Pointwise, small K (16–144 ch)	11.1–11.8 cyc/MAC	6.6–7.0	1.69–1.70 ×
Pointwise, large K (320–960 ch)	13.8–13.9 cyc/MAC	9.2–9.3	1.49–1.50 ×
Depthwise 3×3 (with pair copy)	16.99 cyc/MAC	10.86	1.56 ×
First conv 3×3, stride 2	18.28 cyc/MAC	9.22	1.98 ×
ReLU6 (PRELU6)	14.00 cyc/elem	3.52	3.98 ×

Whole MobileNetV2 + GRU inference (estimate)	Cycles	Speedup
Baseline Shakti	31.9 B	1.00×
+ PFMA (pointwise, depthwise, first conv)	19.8 B	1.61×
+ PRELU6	19.3 B	1.66 ×

Module 5: FPGA Integration (Kria KV260)

Existing prototype (SqueezeNet + TCN, INT8)

- ARM Cortex-A53 (PYNQ / Python) + FPGA conv accelerator
- 128 INT8 MAC lanes at 100 MHz, AXI CDMA, three 1024-bit BRAMs
- Validated bit-exact on the board: 0 mismatches in 788,544 outputs

Measured per frame on the board

Output rebuild (ARM)	1.29 s
Max-pool (ARM)	0.31 s
Accelerator conv	0.18 s
DMA + staging	0.04 s
Total per frame	1.94 s

Plan

Replace the ARM with the Shakti soft core on the FPGA, port the host driver to bare-metal C, and target the CPU-side bottleneck (data reshuffling, pooling, requantisation) with custom instructions.

Module 6: Target Model: TinyMetro-Net (Team 1)

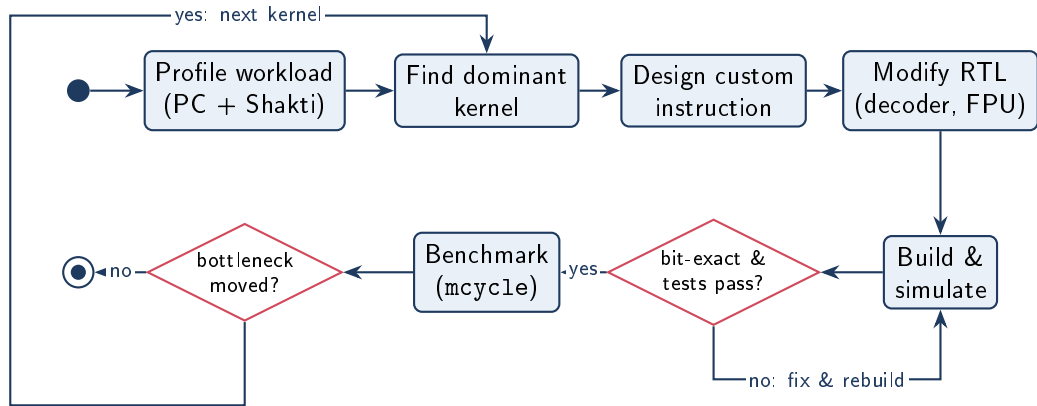
- 13 layers (12 conv + 1 linear), **48,600 INT8 parameters** (47 KB)
- Conv 3×3 + $5 \times$ (depthwise s2 + pointwise) + final pointwise
- Requantisation:
 $out = clip((acc \cdot s) \gg 15, 0, 127)$
- 8 classes (violence, falls, crowd, wrong-way, trespass, luggage, fire, normal)

Where the work is (our calculation)

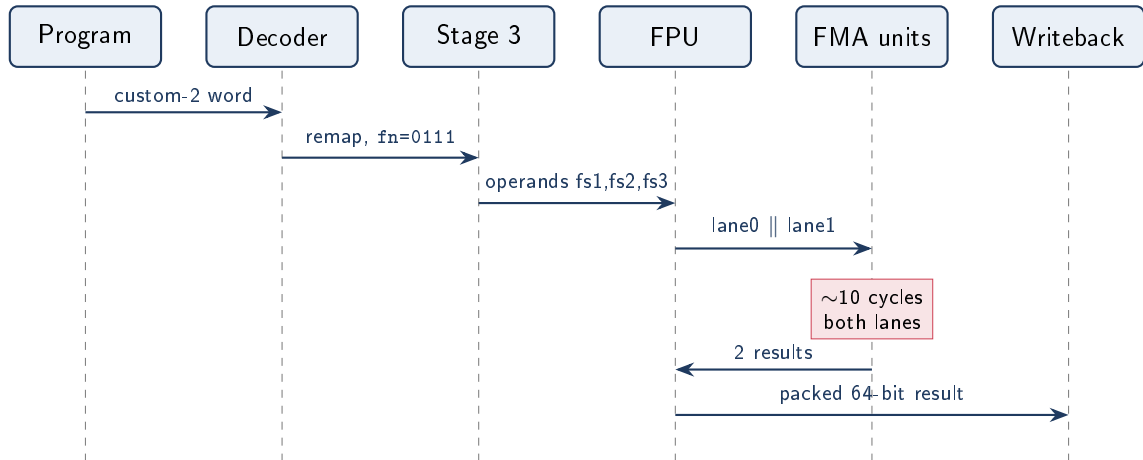
Part	MACs	Share
conv1	8.13 M	57%
Pointwise $\times 6$	5.04 M	35%
Depthwise $\times 5$	1.03 M	7%
Total	14.2 M	

Next: a Shakti profile of TinyMetro, then INT8 instructions: a packed INT8 dot product (PDOT8) and a fused requantise + ReLU.

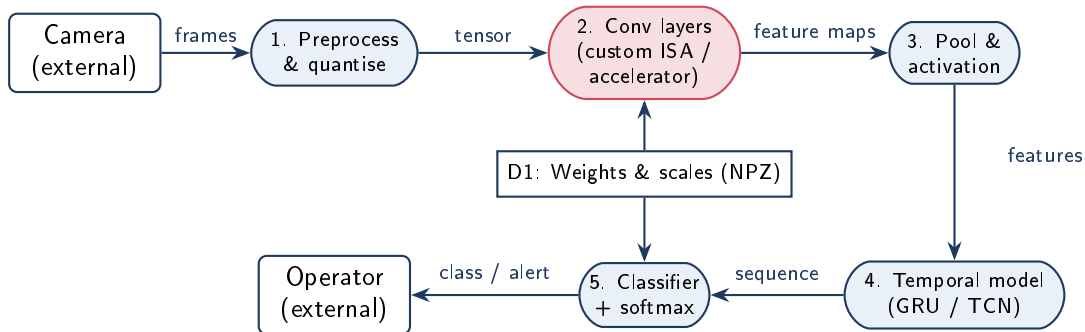
Activity Diagram: Workload-Driven Design Method



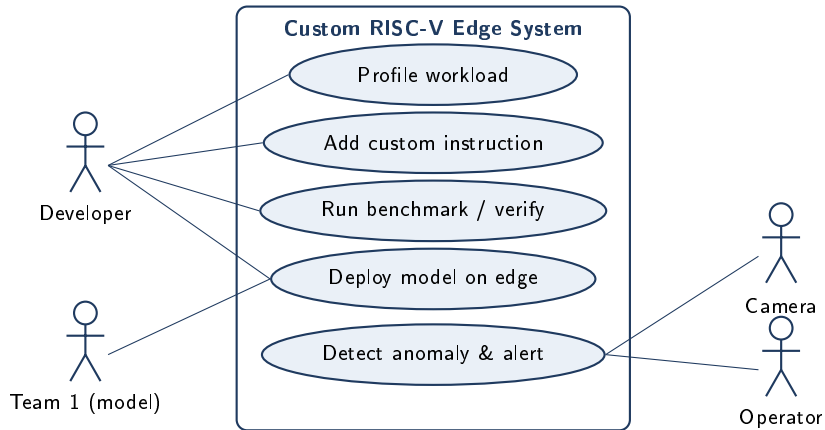
Sequence Diagram: Executing One PFMA.S Instruction



Data Flow Diagram (Level 1): Anomaly Detection Pipeline



Use Case Diagram



Workloads & Test Data

Workload	Data type	Data used
MobileNetV2 + GRU	FP32	ImageNet-pretrained MobileNetV2; 8-frame sequences of 224×224 ; random inputs for timing
SqueezeNet 1.1 + TCN	INT8	Kria package: <code>anomaly_sample.mp4</code> , <code>normal_sample.mp4</code> ; 16 sampled frames
TinyMetro-Net (Team 1)	INT8	8-class metro dataset: 224 test frames, 527 validation frames, 54 held-out clips

- Cycle timing does not depend on the data values, so profiling uses real layer shapes with synthetic data.
- Correctness is checked bit for bit against the baseline implementation.

- **Non-pipelined FPU:** one operation at a time limits the total speedup to $\sim 1.66\times$ (Amdahl's law)
- **Simulation speed:** a full network takes days in Verilator, so we measure sampled layers and extrapolate
- **Memory:** the 16 KB D-cache slows the large layers
- **FPGA integration:** no Vivado project yet; fitting Shakti and the accelerator on the K26 is unverified
- **Software port:** the Python driver must become bare-metal C
- **Data type:** target models are INT8, while our first instructions are FP32
- **Tooling:** stale builds, long rebuilds (15–20 min)

Expected Output: Achieved So Far

Deliverable	Status	Result
PyTorch + Shakti profiling	✓	pointwise = 84% of the time
FDOT2.S	✓	$1.87\times$ fewer instructions, $1.07\text{--}1.13\times$
PFMA.S (2nd FMA unit)	✓	$1.5\text{--}2.0\times$ on conv kernels
PRELU6.S	✓	$3.98\times$ on ReLU6
Regression (riscv-tests)	✓	all pass, bit-exact
Whole-model estimate	✓	31.9 B $\rightarrow$ 19.3 B cycles ($1.66\times$)
Kria prototype analysis	✓	bottleneck = ARM data handling

Expected Output: Final System

- 1 **Pipelined FPU**: a new FMA every cycle (projected pointwise $\sim 2-3$ cyc/MAC, $\sim 3-4\times$ overall; to be measured)
- 2 **INT8 instructions** for TinyMetro-Net: packed dot product, fused requantise + ReLU
- 3 **Complete network** running on the customised Shakti core, compared end to end with baseline Shakti
- 4 **Shakti + accelerator on the Kria KV260**, replacing the ARM core
- 5 **Edge vs server** comparison of latency and throughput

Comparison table (final report)

Baseline Shakti	vs	Shakti + custom ISA	vs	ARM + accelerator	vs	Shakti + accelerator
			vs	Server		

- A **workload-driven** method (profile $\rightarrow$ bottleneck $\rightarrow$ instruction $\rightarrow$ verify $\rightarrow$ measure) produced real, measured gains on an open RISC-V core.
- Three custom instructions (FDOT2, PFMA, PRELU6) were designed, built into Shakti C-Class and verified **bit-exact**, with standard ISA behaviour unchanged.
- Estimated **1.66 $\times$** speedup for MobileNetV2 + GRU; the remaining bottleneck (the non-pipelined FPU) is identified and quantified.
- Next: INT8 instructions for Team 1's TinyMetro-Net, and deploying Shakti with the FPGA accelerator on the Kria KV260.

References I

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Thank You

Questions?